CH 27 — MAGNETIC FORCE

$\vec{F}=q\vec{v}\times\vec{B}$; $|F|=qvB\sin\theta$
$\vec{F}$ wire: $\vec{F}=I\vec{L}\times\vec{B}$; $|F|=BIL\sin\theta$
$\Phi_B=\int B\cdot dA=BA\cos\theta$ $[T\cdot m^2]$

Circular motion

$r=mv/(qB)$; $T=2\pi m/(qB)$; $\omega=qB/m$
$v=qBr/m$; $KE=q^2B^2r^2/(2m)$

★ Velocity selector / Mass spec

★ **Vel sel:** $v=E/B$ ($qE=qvB \leftrightarrow$ undeflected)
★ $V=Ed$ across plates
★ **Mass spec:** $v=\sqrt{2qV/m}$; **$m/q=B^2R^2/(2V)$**
★ **Free-fall into B:** $v=\sqrt{2gh}$; $F=qvB$

★ Magnetic Torque on Loop

$\vec{\mu} = NIA\cdot\hat{n}$ (mag dipole moment); $\hat{n}$ from RHR around I
$\vec{\tau} = \vec{\mu} \times \vec{B}$; $|\tau|=\mu B\sin\theta=NIAB\sin\theta$
$U = -\vec{\mu}\cdot\vec{B} = -\mu B\cos\theta$
★ Stable equilib: $\vec{\mu} \parallel \vec{B}$ ($\theta=0$, U min). Unstable: antiparallel ($\theta=\pi$).

CH 28 — AMPERE'S LAW

$\oint B\cdot dl = \mu_0 I_{encl}$
Biot–Savart: $d\vec{B} = (\mu_0/4\pi)\cdot(I\, d\vec{l}\times\hat{r})/r^2$
Long wire: $B=\mu_0 I/(2\pi r)$
Inside solid wire (r<R): $B=\mu_0 Ir/(2\pi R^2)$
Solenoid: $B=\mu_0 nI$ ($n=N/L$)
Toroid in: $B=\mu_0 NI/(2\pi r)$; **Out:** $B=0$
Ctr of circ loop: $B=\mu_0 I/(2R)$
Moving pt charge: $B=(\mu_0/4\pi)\cdot qv\sin\theta/r^2$
▶ **Multi-I sign:** CCW path → + out; CW → + in

Parallel wires

$F/L=\mu_0 I_1 I_2/(2\pi d)$. Same dir→ATTRACT; opp→REPEL

★ Cheats

★ **Coax** (I_1 in, I_2 out same dir): $a<r<b$: $B=\mu_0 I_1/(2\pi r)$; $r>c$: $B=\mu_0(I_1+I_2)/(2\pi r)$. Opp&equal: B=0 outside.
★ **3 ∥ wires (eq d):** outer $F/L=\mu_0 I^2/(4\pi d)$; middle (sym)=0
★ **Square loop side a near wire** (ctr d, ∥): $F=\mu_0 I_1 I_2 a^2/[2\pi(d^2-a^2/4)]$

CH 29 — FARADAY & INDUCTION

$\varepsilon=-d\Phi_B/dt$; $\oint E\cdot dl=-d\Phi_B/dt$
$\Phi_B=BA\cos\theta$ ($T\cdot m^2$, not Wb!)
Lenz: induced I opposes $\Delta\Phi$

★ EMF when B(t) and/or θ(t) vary

▶ **B(t) varies, θ const:** $\varepsilon=-A\cos\theta\cdot(dB/dt)$
▶ **θ(t) varies, B const:** $\varepsilon=BA\sin\theta\cdot(d\theta/dt)$
▶ **Both vary:** $\varepsilon=-A[\cos\theta\cdot dB/dt - B\sin\theta\cdot d\theta/dt]$
▶ **Rotating coil:** $\varepsilon=NBA\omega\sin(\omega t)$; $\varepsilon_{max}=NBA\omega$

Motional EMF — Slide-wire

$\varepsilon=BLv$; $I=BLv/R$
F to push: $F=B^2L^2v/R$; $P=(BLv)^2/R$
★ Loop entering/exiting B: $I=BLv/R$ during entry & exit (opp dirs); $I=0$ fully inside

★ Induced E from changing B

★ **Solenoid w/ dI/dt:** $r<R_{sol}$: $E=(\mu_0 nr/2)(dI/dt)$; $r>R_{sol}$: $E=(\mu_0 nR^2/(2r))(dI/dt)$; $r=0$: $E=0$
★ **Cyl w/ $B(t)=B_0+B_1 t$** outside: $|E|=B_1A/(2\pi r)$

Displacement current

$I_D=\varepsilon_0(d\Phi_E/dt)$; $j_D=\varepsilon_0(dE/dt)$
Ampere–Maxwell: $\oint B\cdot dl=\mu_0(I_C+I_D)$
★ Charging cap: $r<R$: $B=\mu_0 I_C\cdot r/(2\pi R^2)$; $r>R$: $B=\mu_0 I_C/(2\pi r)$

CH 30 — INDUCTANCE & RLC

$\varepsilon_L=-L(dI/dt)$; $L=N\Phi_B/I$ [H]
Solenoid: $L=\mu_0 N^2A/\ell$; **Toroid:** $L=\mu_0 N^2A/(2\pi r)$
$U=\frac{1}{2}LI^2$; $u_B=B^2/(2\mu_0)$
Mutual ind: $M=N_2\Phi_{21}/I_1$; $\varepsilon_2=-M(dI_1/dt)$

RL Circuit | τ=L/R

▶ **Energizing:** $i(t)=(\varepsilon/R)[1-e^{-Rt/L}]$
▶ **Decay:** $i(t)=I_0e^{-Rt/L}$
▶ $t=0$: $i=0$ (inductor blocks); $t=\infty$: $i=\varepsilon/R$
★ Find L from decay: $L=Rt/\ln(I_0/I)$

LC Oscillation

$\omega_0=1/\sqrt{LC}$; $f=1/(2\pi\sqrt{LC})$
$q(t)=Q\cos(\omega t)$; $i(t)=-Q\omega\sin(\omega t)$
Energy: $U=Q^2/(2C)=\frac{1}{2}LI^2_{max}$

★ Damped LRC (series)

★ $\omega'=\sqrt{1/LC - R^2/(4L^2)}$
★ Amp decay: $A(t)=A_0e^{-Rt/(2L)}$; t to fraction f: $t=-(2L/R)\ln(f)$
★ Critical: $R_c=2\sqrt{L/C}$. $R<R_c$ under; $R>R_c$ over

CH 31 — AC CIRCUITS

$V(t)=V_{max}\cos\omega t$; $I(t)=I_{max}\cos(\omega t-\varphi)$
$V_{rms}=V_{max}/\sqrt{2}$; $I_{rms}=I_{max}/\sqrt{2}$
R: in phase, $V_R=IR$
L: V leads 90°, $X_L=\omega L$, $V_L=IX_L$
C: V lags 90°, $X_C=1/(\omega C)$, $V_C=IX_C$
"ELI the ICE man" — E leads I in L; I leads E in C

Series LRC

$Z=\sqrt{R^2+(X_L-X_C)^2}$; $I=V/Z$
$\tan\varphi=(X_L-X_C)/R$
Resonance: $X_L=X_C$; $\omega_0=1/\sqrt{LC}$; $Z=R$
$P_{avg}=V_{rms}\cdot I_{rms}\cdot\cos\varphi=I^2_{rms}\cdot R$

★ Phasor Diagram Tips

★ V_R along I (in phase); V_L straight up (+90°); V_C straight down (−90°). $\vec{V}_{source}$ = vector sum.
★ $|V_{source}|^2 = V_R^2 + (V_L-V_C)^2$

★ Transformer (ideal)

★ **$V_2/V_1 = N_2/N_1$** (turns ratio)
★ **$I_2/I_1 = N_1/N_2$** (inverse, conserves P)
★ $P_1=P_2 \rightarrow V_1I_1=V_2I_2$ (ideal)
★ Step-up: $N_2>N_1 \rightarrow V\uparrow, I\downarrow$. Step-down: opposite.

CH 32 — EM WAVES

▶ **$E=E_m\sin(kx-\omega t)$; $B=B_m\sin(kx-\omega t)$**
▶ **$E_m/B_m=c$; $E(t)/B(t)=c$**
▶ **$c=1/\sqrt{\mu_0\varepsilon_0}=f\lambda=\omega/k$**
▶ **$k=2\pi/\lambda$; $\omega=2\pi/T=2\pi f$**
▶ **$\vec{S}=(1/\mu_0)(\vec{E}\times\vec{B})$**; $|S|=EB/\mu_0=E^2/(c\mu_0)=cB^2/\mu_0$
▶ **$I=S_{avg}=E_m B_m/(2\mu_0)=\frac{1}{2}c\varepsilon_0E_m^2=E_m^2/(2\mu_0c)$**
Rad pressure: Absorb: $P=I/c$. Reflect: $P=2I/c$
$\vec{E}\times\vec{B}$ → propagation (RHR). Spectrum (low→high f): Radio < μW < IR < Vis(400-700nm) < UV < X < γ
In medium: $v=c/n$ (n=index of refraction)

★ MAXWELL'S 4 EQUATIONS

1. Gauss (E): $\oint E\cdot dA = Q_{encl}/\varepsilon_0$
2. Gauss (B): $\oint B\cdot dA = 0$ (no monopoles)
3. Faraday: $\oint E\cdot dl = -d\Phi_B/dt$
4. Ampere–Maxwell: $\oint B\cdot dl = \mu_0(I_C + \varepsilon_0\, d\Phi_E/dt)$

★ RIGHT-HAND RULE GUIDE

F=qv×B (chg): fingers v, curl to B, thumb=F (flip if q<0)
B from wire: thumb along I, fingers curl in B dir
Ampere loop: fingers along path, thumb = + I dir
Loop μ: fingers curl with I, thumb = μ dir
EM wave: fingers E, curl to B, thumb = propagation
Lenz: Find dΦ dir, induced B opposes it, then RHR for I

★ COMMON PITFALLS

⚠ Φ_B in $T\cdot m^2$ (NOT Wb) per prof
⚠ RC charging is $1-e^{(-t/RC)}$; discharging is $e^{(-t/RC)}$. Don't swap.
⚠ RL: at t=0 inductor BLOCKS (i=0). Cap at t=0 is wire (V=0).
⚠ "Toroid" r is from CENTER (not from wire).
⚠ Cap disconnected → Q const, E unchanged when d changes. Connected → V const.
⚠ cos vs sin in Φ: cosθ when θ from normal; sinθ if from plane.
⚠ In LRC: $V_C+V_L+V_R \neq V_{source}$ (alg). Use phasor sum.
⚠ In Ampere: r is from current axis, not anywhere else.
⚠ Mass spec: V is accelerating voltage, not vel-selector V.
⚠ τ_{RC} = RC, τ_{RL} = L/R — DIFFERENT, easy to swap.
⚠ "Loop fully inside B" → no dΦ → I = 0 (only changing flux induces).
⚠ Direction of induced I: use Lenz, NOT sign in Faraday's eq.

★ TEST 1 EXTRA HELP (Ch 21-23)

Vector decomposition of E from point q at (x,y):
$E_x = (kq/r^2)\cos\theta = kqx/r^3$; $E_y = (kq/r^2)\sin\theta = kqy/r^3$
$|E| = \sqrt{E_x^2+E_y^2+E_z^2}$; $\hat{r}$ sign: a(−)→use $-\hat{r}$; a(+)→use $+\hat{r}$
Solid disk on axis (radius R, σ uniform):
$E_x = 2\pi k\sigma[1 - x/\sqrt{x^2+R^2}]$
Annulus (disk w/ hole, R_in & R_out):
$E_x = 2\pi k\sigma[1/\sqrt{x^2+R_{in}^2} - 1/\sqrt{x^2+R_{out}^2}]$
Ring V on axis: $V = kQ/\sqrt{z^2+a^2}$
Disk V on axis: $V = 2\pi k\sigma[\sqrt{x^2+a^2} - x]$
Ring max E location: at $z = R/\sqrt{2}$ (E=0 at center, max at $R/\sqrt{2}$)
Charge density for integration:
Linear: $dq = \lambda\, dl$ | Surface: $dq = \sigma\, dA$ | Volume: $dq = \rho\, dV$
Semicircle: $dq = (Q/\pi a)\cdot a\, d\theta = (Q/\pi)d\theta$
SHM in E-field (charge q oscillating near ring/disk):
$\omega = \sqrt{kqQ/mR^3}$; $f = (1/2\pi)\sqrt{kqQ/mR^3}$
Conservation of energy (charge moving in E):
$K_a + U_a = K_b + U_b$ | $\frac{1}{2}mv^2 = q\Delta V$ | $v = \sqrt{2q\Delta V/m}$
Work btw two points: $W = kq_1q_2(1/r_1 - 1/r_2)$
ΔV pt charge btw 2 pts: $\Delta V = kQ(1/r_b - 1/r_a)$
E from V (gradient): $E = -dV/dx$; if $V=ar^2 \rightarrow E=-2ar$
Line/cylinder V undefined at ∞ — use ΔV between 2 finite points only

★ TEST 2 EXTRA HELP (Ch 24-27)

★ Force btw cap plates: $F = Q^2/(2\varepsilon_0A) = U/d$ (constant Q)
Cap series/parallel rules:
Series: Q same; V adds ($V_{total} = V_1+V_2$); $1/C_{eq} = \Sigma 1/C_i$
Parallel: V same; Q adds ($Q_{total} = Q_1+Q_2$); $C_{eq} = \Sigma C_i$
Voltage divider (series C): $V_1 = V_{total}\cdot(C_{eq}/C_1)$
Charge divider (parallel C): $Q_1 = Q_{total}\cdot(C_1/C_{eq})$
★ Meter rules:
Ammeter: 0 Ω, in SERIES | Voltmeter: ∞ Ω, in PARALLEL
★ Kirchhoff loop sign rule:
Walk WITH current thru R: −IR | Walk AGAINST current: +IR
Exit + terminal of battery: +ε | Exit − terminal: −ε
Battery power breakdown:
Total $P = \varepsilon I$ | Useful (to R) $P = V_{term}\cdot I = (\varepsilon-Ir)I$
Wasted in r: $P_{lost} = I^2r$ | Energy = Pt
Power absorbed (charging batt): $P = \varepsilon I$ (energy goes IN)
Cap work done (Δd): $W = U_{final} - U_{initial}$
F = qv×B determinant form:
F = q·det\|î ĵ k̂; vx vy vz; Bx By Bz\|
Wire perpendicular to B: $F = ILB$ ($\sin\theta = 1$)
Total charge from N electrons: $q = N\cdot e$ ($e = 1.6\times10^{-19}$ C)
Kinematics in B-field (free fall into B): $v^2 = v_0^2 + 2g\Delta y$
V_xy potential difference: $V_{xy} = V_x - V_y$

🎯 EXAM TRIAGE — KEYWORD → FORMULA

"insulating sphere E in/out" → kQr/R^3 vs kQ/r^2
"infinite line/wire E" → $\lambda/(2\pi\varepsilon_0 r)$
"infinite sheet E" → $\sigma/(2\varepsilon_0)$
"conductor surface E" → σ/ε_0
"point charge E" → kQ/r^2
"point charge V" → kQ/r
"ring on axis E" → $kqz/(z^2+a^2)^{3/2}$
"semicircle ±Q" → $4kQ/(\pi a^2)$
"square alt ±q" → $4\sqrt{2}\, kq/a^2$
"cavity in sphere" → $\rho h/(3\varepsilon_0)$
"flux thru hemisphere" → $q/(2\varepsilon_0)$
"Φ pt charge full sphere" → q/ε_0
"V from line of charge" → $2k\lambda\ln(r_0/r)$
"work to move charge" → $W=q\Delta V$
"speed gained from V" → $v=\sqrt{2qV/m}$
"E between plates" → V/d
"force on charge in E" → $F=qE$
"dipole torque" → $pE\sin\theta$
"dipole U" → $-pE\cos\theta$
"cap C from area" → ε_0A/d
"cap energy" → $\frac{1}{2}CV^2 = Q^2/(2C)$
"cap energy density" → $\frac{1}{2}\varepsilon_0E^2$
"cap with dielectric" → $C=\kappa\varepsilon_0A/d$
"cap disconn., move plates" → Q const, $U\propto d$
"cap stays connected" → V const, $U\propto 1/d$
"caps in series" → $1/C_{eq}=\Sigma 1/C_i$
"caps in parallel" → $C_{eq}=\Sigma C_i$
"two C in series" → $C_1C_2/(C_1+C_2)$
"equiv resistance" → reduce groups
"two R parallel" → $R_1R_2/(R_1+R_2)$
"n equal R parallel" → R/n
"power in resistor" → $I^2R = V^2/R$
"power from battery" → εI
"battery terminal V" → $\varepsilon-Ir$
"two batteries opposing" → $I=|\varepsilon_1-\varepsilon_2|/\Sigma R$
"two batteries aiding" → $I=(\varepsilon_1+\varepsilon_2)/\Sigma R$
"wire R from L,A" → $R=\rho L/A$
"drift velocity" → $I=nqv_d A$
"current density J" → I/A
"τ for RC" → RC
"τ for RL" → L/R
"RC voltage at time t" → $V_C=\varepsilon(1-e^{-t/RC})$
"RC current" → $(\varepsilon/R)e^{-t/RC}$
"RC initial (t=0)" → $V_C=0$, $V_R=\varepsilon$, $i=\varepsilon/R$
"RC final (t=∞)" → $V_C=\varepsilon$, $V_R=0$, $i=0$
"RC discharge V" → $V_0\, e^{-t/RC}$
"time to charge to half" → $t=RC\ln 2$
"time to decay to 1/e" → $t=\tau$
"loop entering B" → $I=BLv/R$
"slide wire EMF" → $\varepsilon=BLv$
"force to push rod" → B^2L^2v/R
"power dissipated rod" → $(BLv)^2/R$
"undeflected ion" → $v=E/B$
"mass spec m/q" → $B^2R^2/(2V)$
"speed after accel V" → $\sqrt{2qV/m}$
"radius circ motion B" → $r=mv/(qB)$
"period in B-field" → $2\pi m/(qB)$
"cyclotron freq" → $qB/(2\pi m)$
"force ∥ wires/length" → $\mu_0I_1I_2/(2\pi d)$
"toroid B inside" → $\mu_0NI/(2\pi r)$
"toroid B outside" → 0
"solenoid B" → μ_0nI
"moving charge B" → $(\mu_0/4\pi)qv\sin\theta/r^2$
"long wire B" → $\mu_0I/(2\pi r)$
"inside wire B (r<R)" → $\mu_0Ir/(2\pi R^2)$
"center loop B" → $\mu_0I/(2R)$
"on axis loop B" → $\mu_0IR^2/[2(R^2+z^2)^{3/2}]$
"torque on loop" → $\mu\times B = NIAB\sin\theta$
"PE of dipole in B" → $-\mu\cdot B$
"dI/dt in solenoid → E" → $(\mu_0nr/2)(dI/dt)$
"charging cap → B" → $\mu_0I_C\, r/(2\pi R^2)$
"displacement current" → $I_D=\varepsilon_0\, d\Phi_E/dt$
"rotating coil ε" → $NBA\omega\sin(\omega t)$
"max EMF rotating" → $NBA\omega$
"flux when B(t)" → $\varepsilon=-A\cos\theta\, dB/dt$
"flux when θ(t)" → $\varepsilon=BA\sin\theta\, d\theta/dt$
"L of solenoid" → μ_0N^2A/ℓ
"L of toroid" → $\mu_0N^2A/(2\pi r)$
"inductor energy" → $\frac{1}{2}LI^2$
"u_B (B-field density)" → $B^2/(2\mu_0)$
"RL energizing" → $(\varepsilon/R)[1-e^{-Rt/L}]$
"RL decay" → $I_0\, e^{-Rt/L}$
"RL find L" → $L=Rt/\ln(I_0/I)$
"LC freq" → $1/(2\pi\sqrt{LC})$
"LC ω₀" → $1/\sqrt{LC}$
"LRC critical R" → $2\sqrt{L/C}$
"LRC underdamped ω" → $\sqrt{1/LC-R^2/4L^2}$
"amplitude decay LRC" → $e^{-Rt/2L}$
"X_L" → ωL
"X_C" → $1/(\omega C)$
"impedance Z" → $\sqrt{R^2+(X_L-X_C)^2}$
"resonance ω" → $1/\sqrt{LC}$
"phase angle" → $\tan\varphi=(X_L-X_C)/R$
"P_avg AC" → $I_{rms}^2\, R$
"V_rms" → $V_{max}/\sqrt{2}$
"transformer V" → $V_2/V_1 = N_2/N_1$
"transformer I" → $I_2/I_1 = N_1/N_2$
"EM wave B from E" → $B=E/c$
"intensity from E_m" → $\frac{1}{2}c\varepsilon_0E_m^2$
"Poynting" → $(1/\mu_0)E\times B$
"|S|" → $EB/\mu_0 = E^2/(c\mu_0)$
"rad pressure absorb" → I/c
"rad pressure reflect" → $2I/c$
"k from λ" → $2\pi/\lambda$
"ω from f" → $2\pi f$
"c from μ₀,ε₀" → $1/\sqrt{\mu_0\varepsilon_0}$
"E and B in phase" → both $\sin(kx-\omega t)$
"Maxwell ∮E·dA" → Q_{encl}/ε_0
"Maxwell ∮B·dA" → 0
"Maxwell ∮E·dl" → $-d\Phi_B/dt$
"Maxwell ∮B·dl" → $\mu_0(I_C+\varepsilon_0\, d\Phi_E/dt)$
"speed in medium" → c/n
"f×λ" → c (or v in medium)
"E density u_E" → $\frac{1}{2}\varepsilon_0E^2$